

Detecting Sexism in Tweets: A Multi-Task Approach

EXIST 2025 Dataset Analysis and Classification

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1 Introduction & Methodology

The EXIST 2025 dataset contains tweets annotated by multiple individuals for sexist content. Our pipeline comprises:

- **Data Aggregation:** Majority voting on multiple labels, removal of ties, and computation of agreement ratios.
- **Text Preprocessing:** Cleaning (HTML, usernames, URLs), tokenization, stopword removal, stemming (quanteda).
- **Feature Engineering:**
 - **Lexical:** Presence of discriminatory terms (e.g., “bitch”, “feminist”), offensive collocations.
 - **Sentiment/Emotion:** NRC Emotion Lexicon, overall sentiment scores, mixed sentiment detection.
 - **Linguistic:** TF-IDF vectors, average word/sentence length, punctuation density, capitalization patterns.
 - **Demographic:** Proportion of annotators by gender, age group, education, ethnicity per tweet.
- **Balancing:** SMOTE oversampling to address class imbalance.
- **Validation:** Stratified 5-fold cross-validation; primary metric: ROC AUC, secondary: Precision, Recall, F1.
- **Test Set:** The `EXIST2025_dev_labeled.csv` file was used as the held-out test set.

2 Task 1: Sexism Classification Models

2.1 Models Evaluated

We trained six models (Logistic Regression, SVM with RBF, Decision Tree, Random Forest, Gradient Boosting Machine (GBM)) with grid-search hyperparameter tuning. Table 1 presents their averaged CV results:

Model	Acc.	F1	Prec.	AUC
Logistic Regression	0.731	0.700	0.713	0.753
SVM (RBF)	0.722	0.678	0.708	0.757
Decision Tree	0.741	0.708	0.725	0.725
Random Forest	0.750	0.708	0.747	0.768
GBM (best)	0.778	0.750	0.770	0.778

GBM (max_depth=6, learning_rate=0.1, n_estimators = 150) outperformed all others, achieving an AUC of 0.778.

3 Task 2: Demographic Bias Analysis

We tested whether annotator demographics influenced labeling rates via chi-squared tests:

- **Gender:** $p = 0.223$ (not significant)
- **Age Group:** $p = 9.5 \times 10^{-11}$ (significant)
- **Country:** $p < 2.2 \times 10^{-16}$ (significant)
- **Education:** $p = 2.0 \times 10^{-9}$ (significant)
- **Ethnicity:** $p = 1.7 \times 10^{-6}$ (significant)

Feature Engineering: `prop_age_18_25`, `prop_doctorate`, `prop_middle_eastern`, `prop_country_X`, etc. **Impact:** Incorporating only the `prop_middle_eastern` feature into the GBM classifier improved AUC from 0.7783 to 0.7882 (+1%).

4 Task 3: Annotator Clustering

To analyze labeling variability, we clustered annotators based on one-hot-encoded demographic attributes (gender, age, education, ethnicity, region). After preprocessing and scaling, K-Means clustering with $k = 4$ was selected using the Calinski-Harabasz index (2891.96) and silhouette score (0.397), indicating moderate cluster quality with reasonable separation and some overlap (Dunn index = 0.071).

Although the clusters revealed certain behavioral patterns, incorporating the corresponding cluster assignments (`cluster_1`, `cluster_2`, `cluster_3`, `cluster_4`) as additional features in the classifier did not improve performance. In fact, AUC dropped to 0.6875 and the F1-score also decreased, suggesting that the cluster information may have introduced noise rather than enhancing generalization.

5 Task 4: Association Rule Mining

We applied the Apriori algorithm (support 0.005, confidence 0.60) on transactions combining tweet tokens and annotator metadata. Key rules associated with sexist labeling included:

- $\{\backslash\text{blond}, \backslash\text{dumb}\} \Rightarrow \text{Label}=\text{YES}$ (conf = 0.92, lift = 2.20)
- $\{\backslash\text{bitch}, \text{Gender}=\text{F}\} \Rightarrow \text{Label}=\text{YES}$ (conf = 0.85, lift = 2.05)
- $\{\backslash\text{cook}, \backslash\text{husband}\} \Rightarrow \text{Label}=\text{YES}$ (conf = 0.88, lift = 2.12)

Non-sexist rules included abstract or professional terms (e.g., $\{\text{Ethnicity}=\text{White}, \backslash\text{ideolog}\} \Rightarrow \text{Label}=\text{NO}$, conf = 0.92). These rules guided the creation of interpretable features such as `contains_whore_and_female` and `contains_bitch_and_male`. However, incorporating these rule-based features into the classifier did not lead to performance improvements.

6 Task 5: Predicting Annotator Behavior

To handle annotation subjectivity, we modeled annotator tendencies using demographic-based estimates and behavioral metrics. These features were designed to contextualize labeling noise and improve model robustness:

- `predicted_sexist_rate`: Demographic-based bias estimate.
- `demographic_knn_bias`: Local average bias from similar annotators.
- `bias_confidence`: Confidence in bias estimation.
- `label_entropy`, `label_variance`: Labeling consistency metrics.

However, among the derived features, only `agg_predicted_sexist` and `agg_mean_label` yielded performance improvements when aggregated at the tweet level.

7 Task 6: Final Integration

We combined the most effective features from prior tasks to refine the GBM classifier.

Model	AUC	F1 (Macro)
Baseline (T1)	0.7783	0.7500
+ Demographics (T2)	0.7882	0.7500
+ Annotator Bias (T5)	0.7974	0.7562
Final (T2+T5)	0.7912	0.7855

Threshold Tuning

Optimizing the classification threshold raised the F1 score from 0.7679 to 0.7855 without impacting AUC.

8 Conclusion

Our multi-task approach shows that modeling annotators is critical for fair and accurate sexism detection. Bias-aware and behavioral features improved classifier reliability and helped mitigate subjectivity in labeling. The final model achieved AUC = 0.7912 and F1 = 0.7855, confirming the value of integrating demographic insights with text-based learning.